Two-dimensional powder diffraction in layered van der Waals films: quantitative X-ray area-detector modeling

1 Introduction

Layered van der Waals solids are often prepared as thin films, where their anisotropic crystal structures make crystallite orientation and phase central to the measured response.[1–3] Depending on the substrate and growth conditions, the crystallites may align their layer normals with the film normal while retaining little common in-plane azimuthal registry.[3] We refer to this uniaxially textured limit as a two-dimensional (2D) powder: individual crystallites remain three-dimensional, but one crystallographic axis is preferentially aligned while their rotations about it are approximately random.[4, 5] The resulting reciprocal-space distribution produces characteristic specular and off-specular features that an area detector records simultaneously, providing access to their detector-space positions, apparent line shapes, azimuthal extents, and intensities for corrected quantitative comparison.[6–8]

The difficulty is that the same detector features can arise from several correlated experimental and sample-dependent causes, including finite wavelength bandwidth, beam divergence, grazing-incidence refraction, sample misalignment, mosaic spread, and disorder.[4, 8–10] One solution is a forward model that starts from a crystallographic and structural hypothesis and propagates it through the incident beam distribution, sample geometry, scattering kinematics, and detector response to predict the area-detector image for comparison with experiment. Keeping the calculation in detector space allows the assumptions that generate each feature to be tested against the measured two-dimensional intensity distribution rather than only a one-dimensional reduction. Figure 1 places the 2D-powder state between the single-crystal and 3D-powder limits in real, reciprocal, and detector space.

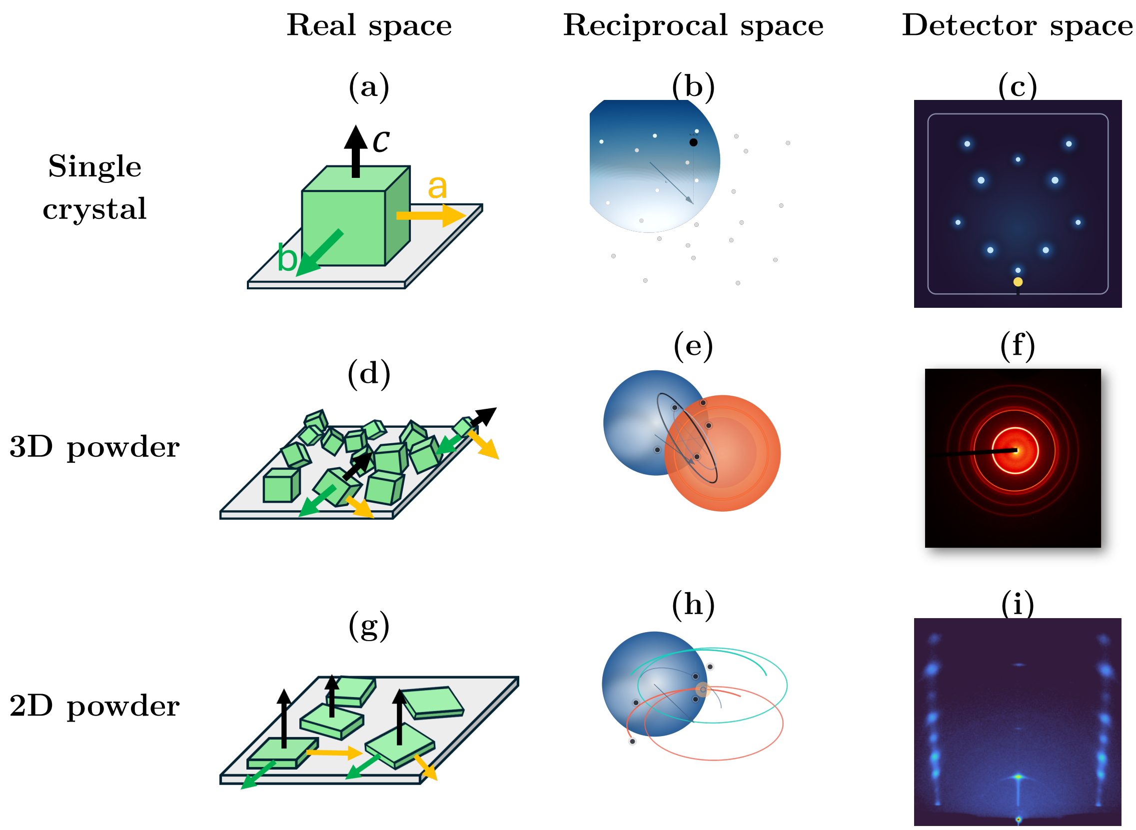


Figure 1: Diffraction across three orientational limits in real, reciprocal, and detector space. In the reciprocal-space panels (b,e,h), the blue Ewald sphere has radius $|\mathbf{k}_i| = 2\pi/\lambda$; dark points or curves mark reciprocal-space intensity intersected by that sphere. (a–c) A single crystal has isolated reciprocal-lattice points, so isolated crossings give sparse detector spots; panel (c) is schematic, and its yellow center marks the direct-beam/beamstop region. (d–f) Random 3D orientations spread a reflection over a spherical shell, whose intersection with the Ewald sphere maps to a Debye–Scherrer ring. (g–i) A 2D powder retains a common film normal but is random in in-plane azimuth. Off-specular reflections become rings about the film-normal reciprocal-space axis; their Ewald-sphere crossings map to paired detector features when both fall within detector acceptance, while $00L$ intensity remains near the specular direction.

Figure 1 summarizes the real-space, reciprocal-space, and detector-space consequences of these three orientational limits. Here $\mathbf{k}_i$ and $\mathbf{k}_f$ are the incident and exit wavevectors, and $\mathbf{G}$ is a reciprocal-lattice vector. With this convention, the Ewald sphere is centered at $-\mathbf{k}_i$, passes through the reciprocal-space origin, and has radius $|\mathbf{k}_i| = 2\pi/\lambda$. Elastic diffraction occurs where reciprocal-space intensity satisfies $|\mathbf{k}_i + \mathbf{G}| = |\mathbf{k}_i|$; each allowed point defines an exit direction $\mathbf{k}_f = \mathbf{k}_i + \mathbf{G}$.[11] For a single crystal, reciprocal space consists of isolated Bragg points [Fig. 1(b)], so only isolated sphere–point crossings produce the sparse detector spots in Fig. 1(c). For a 3D powder, random crystallite orientations spread each reflection over a spherical shell [Fig. 1(e)]; the sphere–shell intersection is a circle that maps to the Debye–Scherrer ring in Fig. 1(f).[11] For a 2D powder, azimuthal disorder about the common c axis spreads each off-specular Bragg point into a ring about the film-normal reciprocal-space axis, while specular ($00L$) intensity remains near that axis [Fig. 1(h)].[4, 6] The Ewald sphere generally cuts an off-specular ring at two points, which map to paired detector features when both directions fall within detector acceptance [Fig. 1(i)].[6, 7, 12] Mosaicity then broadens these ideal intersections into arcs and cap-like intensity rather than changing the basic diffraction condition.[4, 6]

Figure 2 shows a representative measured 2D-powder pattern in which the Bragg peaks, arcs, and streaks retain this anisotropic detector-space intensity distribution.

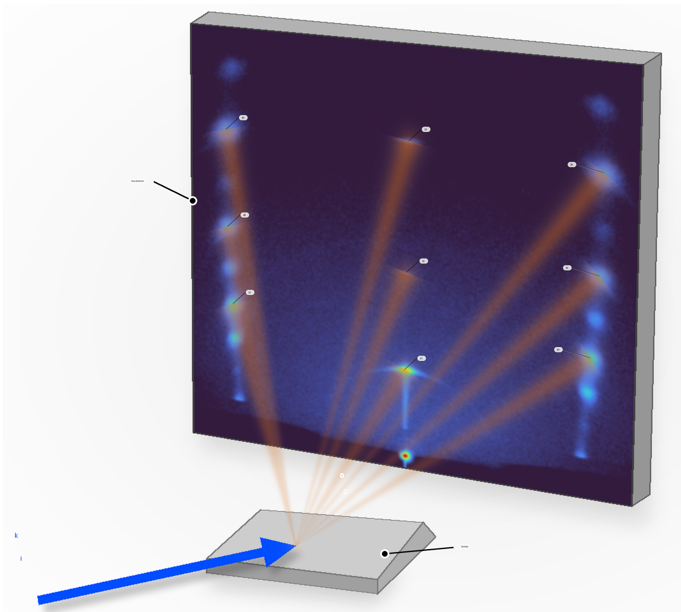


Figure 2: Illustration of a 2D-powder WAXS measurement using a measured PbI_2 sample. The incident beam strikes the film, and diffracted intensity is recorded directly on the area detector. Bright features indicate Bragg peaks, while the streaks between them and arcs illustrate how oriented crystallites, finite mosaic spread, and detector geometry distribute reciprocal-space intensity across the detector image.

This hybrid detector-space information is what makes the 2D-powder problem both rich and difficult. Many standard WAXS workflows reduce area-detector images to 1D profiles by azimuthal integration, often with masking near the direct beam and specular region.[7, 8, 13–15] Those steps suit nearly isotropic powders, but for 2D powders they deweight or remove off-specular arcs and specular caps that constrain out-of-plane correlations, stacking, and mosaic tails.[6, 7] Stacking disorder is also commonly inferred from 1D interference models and refinements,[16–18] which do not directly exploit the two-dimensional anisotropy of oriented-film scattering. More general intermediate orientation distributions can be treated with orientation-distribution-function (ODF) refinements or preferred-orientation corrections,[19–24] but those approaches are more general than needed for the narrowly constrained 2D-powder case considered here.[23, 24] Recent radial-profile approaches have made thin-film GIXD substantially more quantitative, but peak overlap and background separation can remain limiting in strongly anisotropic patterns, especially at small incidence angles.[25] For the layered films considered here, the measured morphology already motivates axial symmetry about the film normal and a low-dimensional out-of-plane mosaicity kernel.[4, 5, 26] We therefore assume this physically motivated 2D-powder orientation distribution a priori and refine it with a detector-space intensity forward model, using off-specular arcs and specular-family line shapes as direct constraints.[6, 7]

Prior work has established the main ingredients needed for such a framework, including grazing-incidence scattering formalisms,[27, 28] reciprocal-space mapping and indexing for oriented thin films,[6–8, 12] detector-space visualization and indexing of GIXS/GIWAXS patterns,[29] and forward calculations of diffraction from textured films.[9] Most directly, Smilgies and Li calculated grazing-incidence diffraction-spot positions in detector space rather than nonlinearly transforming the full detector image into reciprocal space.[29] The present work extends that geometric starting point to the detector intensity by including finite beam phase space, sample and detector geometry, grazing-incidence propagation, mosaicity, detector resolution, optical intensity terms, and structure-factor weighting.

Here we present the resulting detector-space intensity forward model for WAXS from 2D-powder films and apply the same framework to ordered Bi_2Se_3 and Bi_2Te_3 films.[2, 3] At $\theta_i = 5°$, calculated hexagonal rod-family trajectories track the selected measured Bragg-feature locations, and profiles extracted from measured and calculated detector images with corresponding integration bands agree closely in their principal peak positions and apparent line shapes (Fig. 7). Images collected at 5°, 10°, and 15° were included in the refinement, so these comparisons test in-sample fit adequacy rather than held-out predictive performance. The result is a direct detector-space comparison that retains anisotropic information obscured by a conventional one-dimensional reduction.

The paper first develops the diffraction geometry and detector-space intensity calculation, then treats mosaicity and the coupled effects of bandwidth, sample alignment, and near-critical optics. It next presents the ordered Bi_2Se_3 and Bi_2Te_3 test cases, followed immediately by the staged refinement workflow used to construct those comparisons. PbI_2 is treated afterward as a separate stacking-disorder extension that retains the detector-space projection while changing the structure factor along fixed in-plane-momentum trajectories. The discussion closes by relating the intensity calculation to detector-space indexing and structural interpretation.

2 Diffraction Model

The detector-space intensity model begins from the same geometric object used for detector-space indexing: the allowed reciprocal-lattice intersections and their detector coordinates.[29] It then goes beyond predicted spot centers by redistributing the structure-factor intensity of the contributing reflections over mosaic distributions. Sampling the incident beam and crystallite orientations, weighting accepted contributions, and mapping the summed intensity to detector pixels provides matched calculated quantities for detector-feature locations, apparent line shapes, integrated regions, and projected L profiles.

2.1 Diffraction Geometry

We first consider a single incident ray striking a perfectly aligned crystallite, with no beam divergence, no refraction, and no absorption. Let $\mathbf{k}_i$ denote the incident wavevector and $\mathbf{k}_f$ the scattered wavevector. For x-ray wavelength λ, elastic scattering preserves their magnitudes,

$$|\mathbf{k}_i| = |\mathbf{k}_f| = \frac{2\pi}{\lambda}. \tag{1}$$

Equation (1) sets the radius of the Ewald sphere. The momentum transfer $\mathbf{Q}$ is the change between the incident and scattered wavevectors,

$$\mathbf{Q} = \mathbf{k}_f - \mathbf{k}_i. \tag{2}$$

For reflection (h, k, l), $\mathbf{G}_{hkl}$ is the corresponding reciprocal-lattice vector. Diffraction occurs when the momentum transfer matches that vector,

$$\mathbf{Q} = \mathbf{G}_{hkl}, \qquad \mathbf{G}_{hkl} = h\mathbf{a}^* + k\mathbf{b}^* + l\mathbf{c}^*, \tag{3}$$

where $\mathbf{a}^*$, $\mathbf{b}^*$, and $\mathbf{c}^*$ are the reciprocal-lattice basis vectors. Equation (3) is the reciprocal-space Bragg condition. Combining Eqs. (2) and (3) gives the scattered wavevector,

$$\mathbf{k}_f = \mathbf{k}_i + \mathbf{G}_{hkl}. \tag{4}$$

For a fixed $|\mathbf{G}_{hkl}|$, the set of possible reflection orientations forms a Bragg sphere of radius $|\mathbf{G}_{hkl}|$ about the reciprocal-space origin. The allowed exit vectors are found by intersecting that Bragg sphere with the Ewald sphere and then propagating the resulting exit ray to the detector.

Incident rays are Monte Carlo samples of beam position, propagation direction, and wavelength. Let a sampled ray be $\mathbf{r}(u) = \mathbf{r}_0 + u\hat{\mathbf{s}}_i$, where $\mathbf{r}_0$ is its origin and $\hat{\mathbf{s}}_i$ is a unit propagation direction. For a sample entrance plane through $\mathbf{r}_s$ with outward unit normal $\hat{\mathbf{n}}$, the forward intersection is

$$u_{\mathrm{hit}} = \frac{(\mathbf{r}_s - \mathbf{r}_0)\cdot\hat{\mathbf{n}}}{\hat{\mathbf{s}}_i\cdot\hat{\mathbf{n}}}, \qquad \mathbf{r}_{\mathrm{hit}} = \mathbf{r}_0 + u_{\mathrm{hit}}\hat{\mathbf{s}}_i, \tag{5}$$

provided the denominator is nonzero and $u_{\mathrm{hit}} > 0$. With orthonormal in-plane sample directions $\hat{\mathbf{e}}_1$ and $\hat{\mathbf{e}}_2$, a ray contributes only when

$$\xi = (\mathbf{r}_{\mathrm{hit}} - \mathbf{r}_s)\cdot\hat{\mathbf{e}}_1, \quad \upsilon = (\mathbf{r}_{\mathrm{hit}} - \mathbf{r}_s)\cdot\hat{\mathbf{e}}_2, \qquad |\xi| \le \frac{W}{2}, \quad |\upsilon| \le \frac{H}{2}, \tag{6}$$

where W and H are the physical in-plane sample dimensions. The footprint is the distribution of accepted hit points and their beam weights; it is not a second displacement applied after the intersection. Applying the incidence geometry means expressing the accepted ray in the sample basis and evaluating its local grazing angle, $\theta_i^{(\mathrm{ray})} = \sin^{-1}(-\hat{\mathbf{s}}_i\cdot\hat{\mathbf{n}})$. Refraction, transmission, and attenuation are then separate optical operations on that accepted ray, as described in Sec. 2.4 and the Supporting Information.

Reflection construction is deterministic rather than Monte Carlo. Integer Miller candidates are enumerated over configured bounds that cover the detector-accessible reciprocal-space range, and systematic absences are imposed by the selected structural model. Monte Carlo sampling does not create Miller indices: it integrates the incident-beam variables and, in a separate draw, the crystallite orientation from the mosaic distribution for each retained candidate. An exit ray satisfying Eq. (4) is propagated to the calibrated detector plane; contributions outside its active area are rejected. The Supporting Information gives this operation order, the sample and detector intersection equations, the symbolic reflection bounds, and the convergence criterion. Numerical bounds, event counts, seeds, and beam-distribution parameters remain configuration inputs that must accompany a reproducible calculation.

In the detector panels, a reflection label identifies the physical (h, k, l) member or the m-indexed in-plane family defined below; a plus or minus sign only distinguishes the two detector-side intersections of that family. Profile extraction means applying a finite detector-space band around a calculated trajectory to both the measured and calculated images, summing signal and normalization over the same pixels, and reporting the result versus Q_z or L. It is therefore a matched projection, not a separate peak-identification algorithm. Figures 3 and 4 summarize the laboratory, detector, and sample-coordinate conventions.

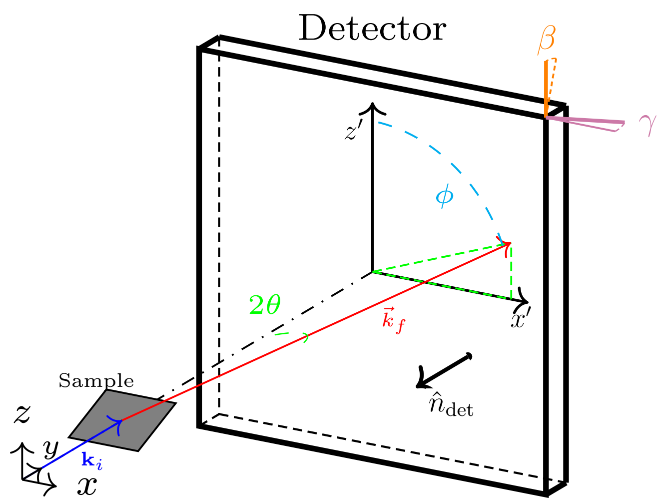
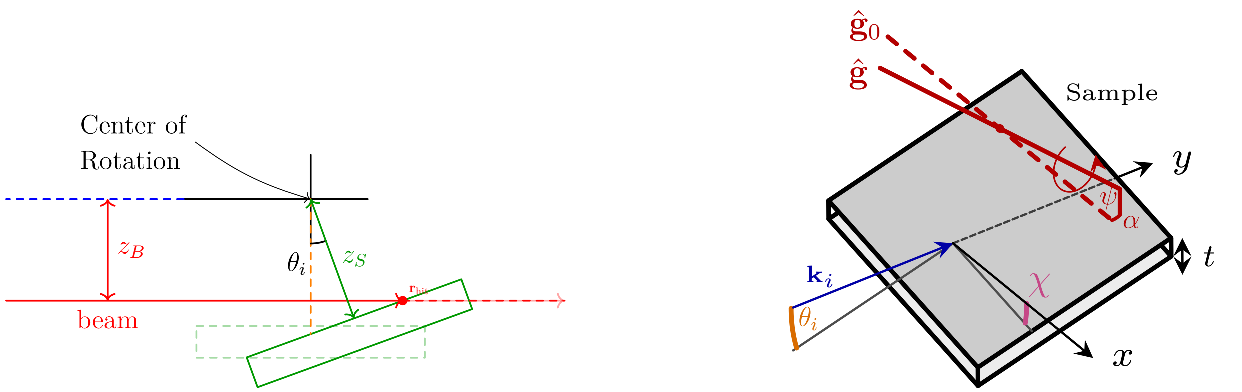


Figure 3: Overall lab-frame geometry. The incident wavevector $\mathbf{k}_i$ strikes the sample, and a representative exit wavevector $\mathbf{k}_f$ reaches the detector at scattering angle 2θ and detector-plane azimuth ϕ. The detector at distance D is tilted by β and γ.



(a) (b)

Figure 4: Sample position and alignment geometry. (a) The red incident ray intersects the film at $\mathbf{r}_{\text{hit}}$. The quantities z_B and z_S are the beam and sample-surface offsets from the goniometer center, and θ_i is the incidence angle; dashed outlines show the zero-angle reference. (b) The incident wavevector is $\mathbf{k}_i$, x and y are laboratory axes, t is the film thickness, and χ and θ_i describe sample rotations about y and x, respectively. The dashed $\hat{\mathbf{g}}_0$ and solid $\hat{\mathbf{g}}$ are the nominal and misaligned goniometer-axis directions; ψ and α rotate that direction about laboratory y and then x, respectively.

2.2 Structure Factor

The reciprocal basis in Eq. (3) includes the factor 2π: if $(\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3) = (\mathbf{a}, \mathbf{b}, \mathbf{c})$, then $\mathbf{a}_i^* \cdot \mathbf{a}_j = 2\pi\delta_{ij}$. An atom with fractional coordinates (x_a, y_a, z_a) therefore has the Cartesian position

$$\mathbf{r}_a = x_a\mathbf{a} + y_a\mathbf{b} + z_a\mathbf{c}. \tag{7}$$

For reflection (h, k, l), the structure factor is[30]

$$F_{hkl} = \sum_a o_a\, f_a(Q)\, \exp(i\mathbf{G}_{hkl} \cdot \mathbf{r}_a)\, \exp\left[-\frac{1}{2}\mathbf{G}_{hkl}^\mathsf{T}\mathbf{U}_a^{\mathrm{cart}}\mathbf{G}_{hkl}\right], \tag{8}$$

where $Q = |\mathbf{Q}| = |\mathbf{G}_{hkl}|$, o_a is occupancy, $f_a(Q)$ is the atomic form factor, and $\mathbf{U}_a^{\mathrm{cart}}$ is the Cartesian mean-square displacement tensor in units of length squared, expressed in the same Cartesian basis as $\mathbf{G}_{hkl}$. If a crystallographic input supplies displacement parameters in a fractional or CIF tensor convention, they must be transformed to $\mathbf{U}_a^{\mathrm{cart}}$ before Eq. (8) is evaluated. The executable transformation is not present in this repository, so this equation defines the required calculation interface. If form-factor tables use $s = \sin\theta/\lambda$, then $s = Q/(4\pi)$. Equation (8) uses one convention throughout: the 2π factor is in $\mathbf{G}_{hkl}$, $\mathbf{r}_a$ is Cartesian, and no second 2π is inserted in the phase. The intensity weight of an allowed member is proportional to $|F_{hkl}|^2$. Symmetry constraints are retained during refinement, so only independent crystallographic parameters are varied.

For the film-normal unit vector $\hat{\mathbf{n}}$, the projected coordinates used for detector-space profiles are

$$Q_z = \mathbf{Q} \cdot \hat{\mathbf{n}}, \qquad \mathbf{Q}_\parallel = \mathbf{Q} - Q_z\hat{\mathbf{n}}, \qquad Q_R = |\mathbf{Q}_\parallel|, \qquad L = \frac{Q_z c}{2\pi}. \tag{9}$$

For an ideally aligned (h, k, l) Bragg vector, $L = l$. Away from an exact Bragg point, L is a continuous projected coordinate rather than a Miller index.

Reflection-family grouping is applied after the structure factor is defined. For the hexagonal basal plane, integer in-plane indices (h, k) are assigned the radial family index

$$m = h^2 + hk + k^2, \tag{10}$$

with in-plane reciprocal-space radius

$$Q_R = \frac{4\pi}{\sqrt{3}\,a}\sqrt{m}. \tag{11}$$

Here a is the hexagonal basal-plane lattice parameter. Thus m labels a radial rod family; it is not a multiplicity.

$$\mathcal{F}_{m,l} = \left\{(h, k) \in \mathbb{Z}^2 : h^2 + hk + k^2 = m,\ (h, k, l) \text{ allowed}\right\}, \qquad \nu_{m,l} = |\mathcal{F}_{m,l}|. \tag{12}$$

The set $\mathcal{F}_{m,l}$ contains all allowed integer members at the same radial index, and $\nu_{m,l}$ is its member count. It is not necessarily one symmetry orbit and is therefore not, in general, a crystallographic multiplicity. Crystallographic multiplicity instead counts the symmetry-equivalent allowed members in one orbit. Detector acceptance and sample orientation can distinguish members of $\mathcal{F}_{m,l}$ even when they share m and Q_R. The detector-space sum is defined over the accepted members individually, so it does not apply a second multiplicity factor after those members have been summed. Figures use m as the compact radial-family label and give explicit (h, k, l) indices when space permits; Q_R is the corresponding physical radial coordinate.

2.3 Mosaicity

Mosaicity is the out-of-plane orientational spread of the layered crystallites about the mean film normal illustrated in Fig. 1(g–i). It is distinct from the random in-plane rotation that defines the two-dimensional powder limit. Random in-plane rotation generates an m-indexed rod family at a common in-plane reciprocal-space radius Q_R, whereas mosaicity spreads each member over crystallite tilt. On an area detector, those two operations appear differently: random azimuthal rotation generates paired off-specular arcs for $m \neq 0$, while out-of-plane tilt broadens the arcs and distributes $m = 0$ intensity away from the ideal specular direction.

Figure 5 shows a full two-dimensional area-detector image of Bi_2Se_3 at $\theta_i = 5°$ with the low-L, $m = 0$ region expanded. At a fixed incident orientation, an ideal perfectly aligned crystallite can satisfy at most one discrete $(00l)$ specular condition, and often none. The measured image instead contains intensity at the nominal 003 and 006 detector locations, near 6° and 12° in the displayed scattering angle 2θ. Their azimuthal extent along the detector-plane angle ϕ is consistent with a distribution of layer-normal tilts. The displayed 2θ values cannot, however, be subtracted from θ_i to obtain the required crystallite tilt; that comparison requires the experimental wavelength and the full vector alignment geometry. The bright near-origin intensity lies in the near-critical specular region and is treated separately in Sec. 2.5; it is not used here as evidence for mosaicity. Bandwidth, divergence, alignment, background, and mosaicity remain coupled in the breadth and visibility of the higher-L features.

Detector-image features of Bi_2Se_3

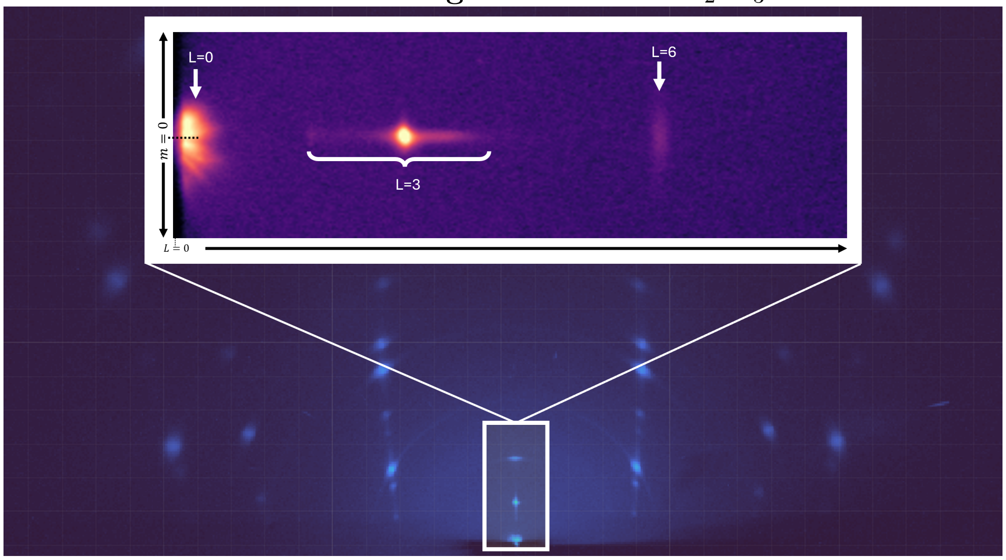


Figure 5: Close-up of the low-L, $m = 0$ detector region from the 5° Bi_2Se_3 image. The bright near-origin intensity lies in the near-critical specular region and is treated separately in Sec. 2.5. Intensity appears at the nominal 003 and 006 detector locations and extends along detector-plane azimuth ϕ. An ideal perfectly aligned crystallite at one incident orientation can satisfy at most one discrete $(00l)$ condition. The coexistence of these features therefore motivates testing an orientational distribution broader than the narrow core, but bandwidth, divergence, alignment, and background must be held fixed before the broad component can be called necessary.

The in-plane rod-family arcs primarily constrain the narrow part of the orientational distribution, whereas weak off-condition specular-family intensity is more sensitive to its broad part. We represent those two scales with the compact phenomenological density

$$\omega_{\mathrm{w}}(\Delta\theta; p_{\mathrm{tail}}, \Gamma, \sigma) = p_{\mathrm{tail}} L_{\mathrm{w}}(\Delta\theta; \Gamma) + (1 - p_{\mathrm{tail}}) G_{\mathrm{w}}(\Delta\theta; \sigma), \qquad 0 \leq p_{\mathrm{tail}} \leq 1. \tag{13}$$

Here G_{w} is the Gaussian-like core, L_{w} is the Lorentzian-like tail, p_{tail} is the tail weight, and $\Delta\theta$ is the crystallite tilt relative to the mean layer normal. The two widths are independent with shared centers. The Lorentzian-like form is a convenient broad-tailed parameterization, not a claim that the physical distribution is uniquely Lorentzian. The present overlays establish that the two-component form is adequate within the refinement set; necessity requires both a fixed-parameter no-tail counterfactual and a fairly re-optimized Gaussian-only comparison.

Figure 6 summarizes the reciprocal-space construction for $m = 0$ and $m \neq 0$. For $m \neq 0$, random in-plane rotation and out-of-plane mosaicity produce off-specular arcs. For $m = 0$, the same angular distribution produces cap-like $00L$ intensity near the specular direction. The radial coordinate is folded onto $Q_R \geq 0$, so opposite signed in-plane tilt components map to the same Q_R locus.

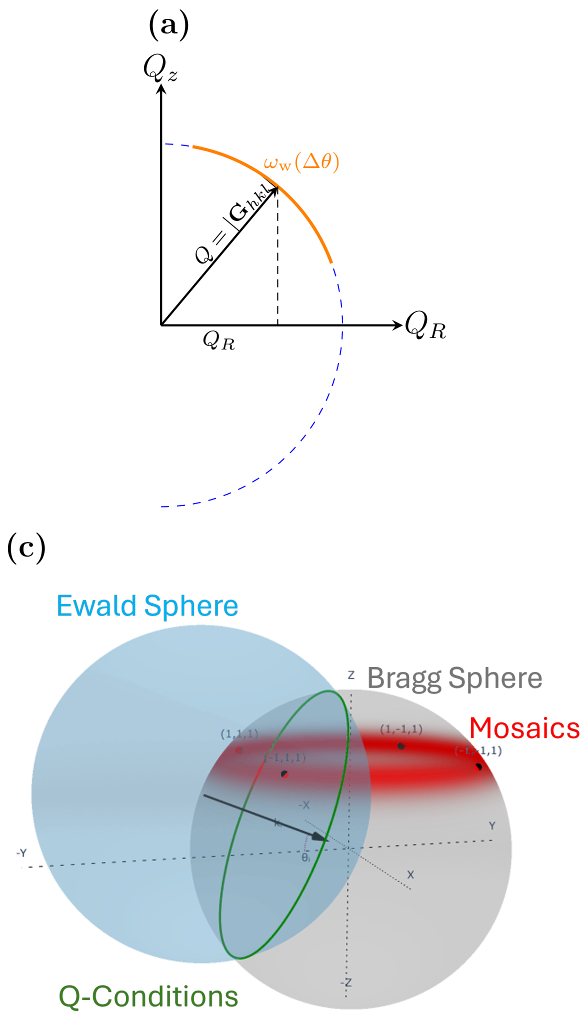


Figure 6: Mosaic broadening and Ewald selection in reciprocal space. (a) For $m \neq 0$, the reciprocal vector has radial and film-normal components Q_R and Q_z; its mosaic distribution occupies an arc on the folded $Q_R \geq 0$ Bragg locus. (b) A specular $m = 0$ vector lies at $Q_R = 0$, and opposite signed tilts fold onto the same radial arc. (c,d) Three-dimensional reciprocal-space views place the corresponding off-specular and specular orientation distributions relative to the Ewald-sphere intersection. The narrow Gaussian-like core and weak Lorentzian-like broad component are the two fitted scales in Eq. (13); these schematics illustrate the selection geometry but are neither simulated detector images nor an ablation test of either component.

2.4 Wavelength sampling and grazing-incidence optics

The low-L, $m = 0$ intensity lies where source bandwidth, crystallite orientation, and grazing-incidence optics are correlated. The present comparison does not isolate those contributions, so we do not assign the feature to bandwidth or optical constants alone. Instead, the forward-model interface treats wavelength sampling and the optical boundary calculation as separate, explicit operations.

For each sampled wavelength λ_j with spectral weight w_j,

$$k_0(\lambda_j) = \frac{2\pi}{\lambda_j}, \qquad \mathbf{k}_i(\lambda_j) = k_0(\lambda_j)\hat{\mathbf{s}}_i, \qquad |\mathbf{k}_i(\lambda_j) + \mathbf{G}| = k_0(\lambda_j). \tag{14}$$

With the reciprocal-space origin fixed, the Ewald sphere for that sample is centered at $-\mathbf{k}_i(\lambda_j)$ and has radius $k_0(\lambda_j)$.[11] Changing wavelength therefore moves the center and changes the radius; the wavelength samples are not concentric shells about one shared center. Each wavelength sample must be tested with the full vector condition in Eq. (14), including the sampled incident direction and crystallite orientation. We consequently make no universal inverse-order bandwidth-scaling claim.

For a layer q, the wavelength-dependent optical input is

$$\tilde{n}_q(\lambda_j) = 1 - \delta_q(\lambda_j) + i\beta_q(\lambda_j). \tag{15}$$

Using the field convention $\exp[i(\mathbf{k}\cdot\mathbf{r} - \omega t)]$ and a depth coordinate $z \geq 0$ directed into the film, phase matching at a flat interface conserves the tangential component and fixes the normal component through

$$\mathbf{k}_{\parallel,q} = \mathbf{k}_{\parallel,0}, \qquad k_{z,q} = \sqrt{[\tilde{n}_q(\lambda_j)k_0(\lambda_j)]^2 - \left|\mathbf{k}_{\parallel,0}\right|^2}. \tag{16}$$

The square-root branch is chosen to decay in the propagation direction; for the inward solution this is $\operatorname{Im} k_{z,q} \geq 0$.[28] The incident and reciprocal exit fields are evaluated separately. The decrement δ_q controls refraction and critical-angle behavior, whereas β_q controls absorption and penetration depth. Both enter every wavelength sample through Eq. (16); whether they materially affect a particular detector region is a sensitivity question, not an assumption tied only to $L \approx 0$.

If $\kappa_i(\lambda_j)$ and $\kappa_f(\lambda_j)$ are the nonnegative normal-depth attenuation constants for the incident and exit fields, scattering from depth z carries the intensity factor

$$W_{\text{depth}}(z; \lambda_j) = \exp[-2\left(\kappa_i(\lambda_j) + \kappa_f(\lambda_j)\right) z]\,. \tag{17}$$

Thus the imaginary normal wavevector is integrated over normal depth, not multiplied by an oblique ray-path length. Interface transmission coefficients are field amplitudes; a transmitted-flux interpretation additionally requires the appropriate normal-flux factor. The wavelength-resolved detector images are then accumulated incoherently,

$$I_{\text{det}}(x, y) = \sum_j w_j I_{\text{det}}[x, y; \lambda_j, \tilde{n}(\lambda_j)]\,. \tag{18}$$

The repository does not yet contain the measured spectrum, the optical-constant table used by the calculation, the executable forward-model source, or direct arrays for the low-L comparison. We therefore do not present an optical-constant ablation or claim that wavelength-dependent optics explain the observed feature. The required test is a full-versus-fixed-$\tilde{n}$ comparison over declared detector ROIs and Q_z ranges; setting $n = 1$ is a separate total-optics-off diagnostic. The Supporting Information records the boundary-condition and evidence requirements without supplying missing numerical inputs.

2.5 Reflectivity

The bright near-origin $m = 0$ intensity in Fig. 5 lies in the near-critical specular region. Its mechanism is distinct from the orientational broadening used to describe the 003 and 006 features. Near the critical angle, refraction, evanescence, absorption, and multiple reflection at the film interfaces can control the intensity, and a Parratt recursion is the appropriate optical limit.[31] The near-origin feature is therefore excluded from the mosaic-tail inference rather than treated as detector-space evidence for tilted crystallites.

The available near-critical comparison is not yet quantitative evidence for a particular optical-to-kinematic handoff. Its archived traces were reconstructed from an annotated raster, the complete blended prediction is absent, and the archive does not establish whether the optical branch is averaged over the mosaic distribution. We therefore omit that provisional overlay from the main text. A quantitative assignment requires direct measured and calculated arrays on matched axes, including the complete prediction and an explicit statement of how crystallite orientation enters the optical branch. Until those inputs are available, the near-origin assignment remains provisional and is kept separate from the mosaicity conclusion.

3 Test cases: Bi_2Se_3 and Bi_2Te_3 films

Ordered Bi_2Se_3 and Bi_2Te_3 films provide test cases for the detector-space intensity calculation. Detector images collected at incident angles of 5°, 10°, and 15° were all used in the refinement; none of these angles is a held-out prediction. The 5° condition is shown because it gives the clearest single-view display of both the off-specular arc families and the specular $m = 0$ features. The comparison therefore tests in-sample agreement in selected feature locations and apparent profile line shapes after the two-dimensional powder construction, geometry, beam phase space, mosaicity, resolution, and ordered structure-factor terms have been applied. It does not by itself establish a unique mosaic decomposition or quantitative relative-intensity accuracy.

The detector-space panels and profile panels in Fig. 7 use the same extraction convention. In panels (a) and (b), the solid colored curves are the calculated centerlines of selected m-indexed Bragg-rod trajectories on the detector. The semi-transparent colored bands are the finite detector regions integrated to form the one-dimensional profiles. The dashed colored outlines mark the edges of those integration bands. The central blue band is the $m = 0$ specular-family trajectory, while the paired off-specular bands are the two detector-side intersections of the same two-dimensional powder rod family. The plus and minus labels identify those two detector-side branches only.

Panels (c) and (d) show the corresponding line profiles after applying the same projection and integration operation to the measured and calculated images. The horizontal coordinate is L, so a marker labeled $L = l$ denotes the nominal $(00l)$ or corresponding off-specular Bragg condition for that selected rod family. The black solid curves are the measured profiles. The orange dashed curves are the simulated profiles after the same detector-space projection. The off-specular rows are plotted on a linear intensity scale, while the $m = 0$ row is plotted on a logarithmic scale because the important result is the persistence of weak, unexpected $00L$ intensity far from the strongest specular peak.

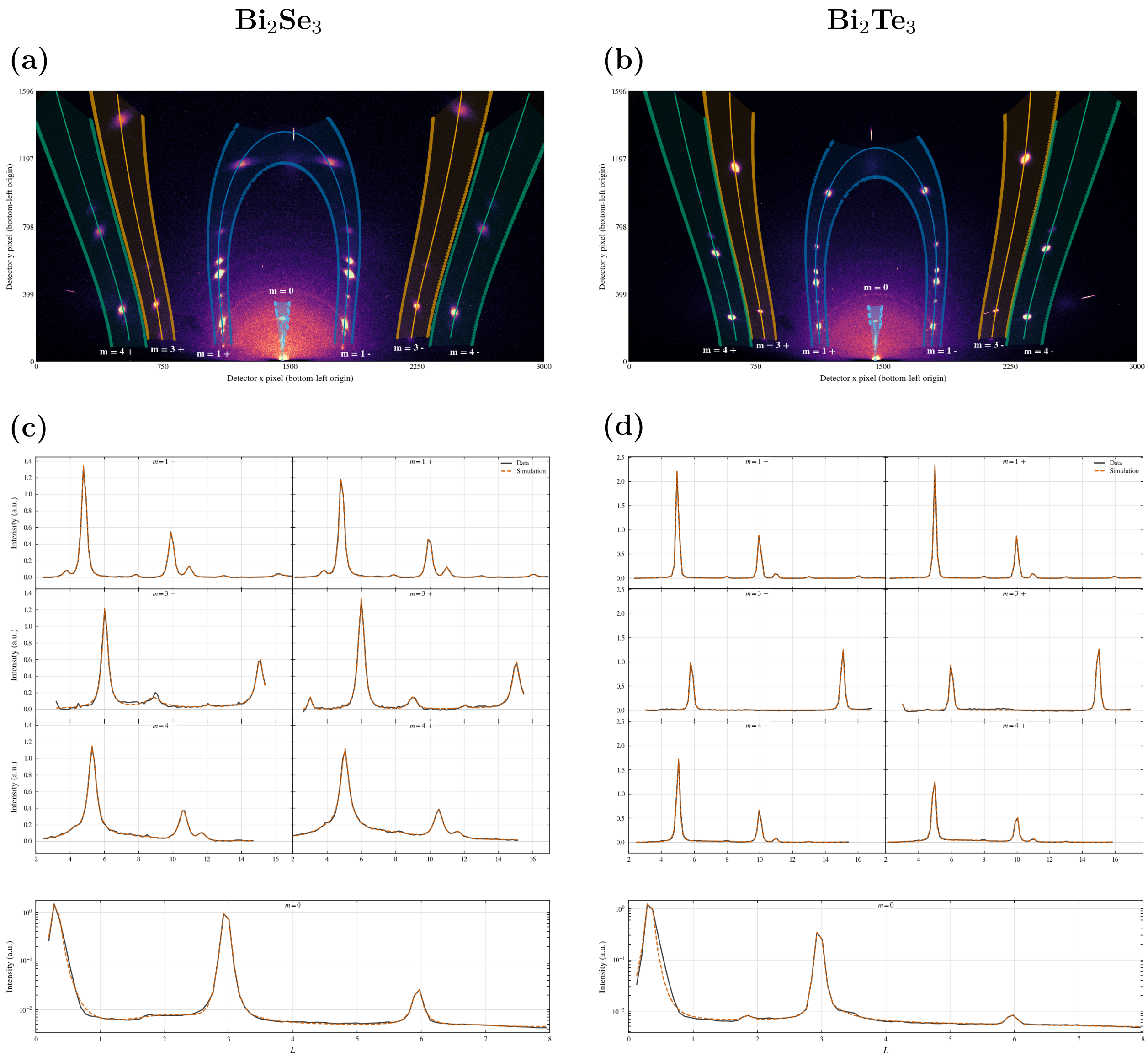


Figure 7: Ordered-film test cases for Bi_2Se_3 and Bi_2Te_3 at $\theta_i = 5°$. (a,b) Measured detector images with selected reciprocal-space extraction regions overlaid. Solid colored curves are trajectory centerlines, semi-transparent regions are the detector pixels integrated to form the profiles, and dashed outlines mark the integration-band boundaries. The labels m identify two-dimensional powder rod families; the signs distinguish the two detector-side intersections of one family. The bright near-origin intensity lies in the near-critical specular region and is treated separately in Sec. 2.5. (c,d) Profiles obtained by applying the same projection to measured and calculated images. Black solid curves are measured data and orange dashed curves are calculations. The off-specular rows compare feature locations and apparent line shapes after geometry, mosaic-core, and resolution terms are applied. The semilogarithmic $m = 0$ rows expose weak off-condition intensity but do not, without a matched ablation, establish that one tail function is uniquely required. All 5°, 10°, and 15° images entered the refinement, so the displayed panels are in-sample comparisons.

Because mosaicity, beam divergence, wavelength spread, incidence-angle uncertainty, detector resolution, and finite integration width all broaden the effective scattering condition, the projected coordinate should not be interpreted as an ideal reciprocal-space cut from a single perfect crystallite. The comparison remains direct because the calculated image is subjected to the same projection and resolution effects before it is compared with the measured profile. This distinction is especially important for the unexpected $00L$ peaks and the enhanced low-L specular intensity, where incidence angle, wavelength bandwidth, divergence, detector-coordinate uncertainty, background, and the broad mosaic component are strongly coupled.

In Fig. 7, the off-specular and $m = 0$ profiles play complementary roles. The off-specular arcs check the two-dimensional powder reciprocal-space construction and calibrated detector geometry, while their widths and asymmetries constrain the narrow mosaic component and instrumental resolution. The off-condition $m = 0$ peaks are sensitive to the broad mosaic component, but their interpretation is also coupled to bandwidth, divergence, background, and alignment. The present overlays show that the fitted model can represent strong and weak detector features within the refinement set; a controlled Gaussian-only comparison is still needed before the broad component can be called necessary or uniquely Lorentzian.

4 Refinement workflow

The refinement is staged rather than fully global from the beginning. Many parameters affect similar detector-space observables: detector distance shifts peak positions, sample alignment changes the apparent incident condition, mosaicity changes profile widths and off-condition intensity, and structure factors change relative peak heights. A simultaneous fit can therefore hide physical errors by compensating one parameter group with another. The staged workflow fixes the instrumental quantities first, then introduces sample orientation, mosaicity, ordered intensity, and finally stacking disorder where needed.

The first distinction is between instrumental parameters and sample parameters. Instrumental parameters describe the beam and detector and should be shared across samples collected under the same experimental configuration. Sample parameters describe the film: incidence condition, sample offset, mosaic distribution, structural intensity, and, for PbI_2, stacking disorder. The refinement sequence in Table 1 uses the detector-space observable most sensitive to each group before that group is used in the final image projection.

The instrumental stages fix where an ideal reflection should appear before any sample-specific physics is refined. Direct-beam images define the incident phase space, including beam width and angular divergence. A powder calibrant fixes the detector mapping: distance, beam center, and detector tilts. These two stages are intentionally separated from sample fitting because an incorrect detector geometry can otherwise be absorbed into apparent sample tilt, mosaic broadening, or shifted reciprocal-space trajectories.

The alignment stage is the part of the workflow most similar to detector-space indexing.

Table 1: Staged refinement workflow. Steps 1–5 are used for the ordered Bi_2Se_3 and Bi_2Te_3 test cases. Step 6 is the added PbI_2 extension for stacking-fault diffuse scattering. All sample images are dark-subtracted before the listed observables are formed, and the final measured/calculated comparison applies the same detector-space projection to data and calculation.

Step	Stage	Data	Parameters refined	Primary information
1	Beam characterization	Direct-beam images at several detector distances	Beam widths w_x, w_z and divergences $\Delta\theta_x$, $\Delta\theta_z$	Incident phase space
2	Detector geometry calibration	Powder calibrant image	Detector distance D, detector tilts (β, γ), and beam center (x_0, y_0)	Detector mapping
3	Sample alignment and goniometer misalignment	Indexed reflection centers from sample images	Sample geometry (θ_i, z_B, z_S), in-plane rotation χ, goniometer misalignment (α, ψ)	Detector-space peak positions
4	Mosaicity/profile refinement	Selected local detector ROIs, specular-tail checks, and local profile shapes	Gaussian-plus-Lorentzian mosaic parameters	Center-aligned profile shapes, off-specular arcs, and specular tails
5	Ordered-film comparison	Selected detector-space Bragg regions and Q_z projections from fixed-m trajectories	Image scale factors and ordered structure-factor parameters	Selected Bragg-feature locations and apparent line shapes
6	PbI_2 diffuse extension	Fixed-Q_R diffuse regions, ordered-baseline residuals, and projected Q_z profiles	Stacking-fault probabilities or layer-pair correlation parameters	Diffuse intensity along Q_z and separation of ordered and disorder-derived scattering

Indexed sample reflections are used to determine how the film sits in the beam, fixing the nominal incident angle, sample offsets, in-plane rotation, and goniometer misalignment well enough that calculated Bragg-rod trajectories can be overlaid on the detector. This establishes where allowed reflections should appear, analogous to detector-space indexing tools such as indexGIXS.[29] Only after this geometric step does it make sense to refine the mosaic distribution, because mosaicity should explain profile shape and off-condition intensity rather than compensate for a misplaced trajectory.

The mosaicity stage constrains the shared narrow-core plus broad-tail angular distribution. Local off-specular profiles constrain the narrow core, while the specular-family profiles are sensitive to the broad component together with bandwidth, divergence, alignment, and background. The ordered-film comparison then applies the same projection to measured and calculated images and compares profiles along fixed-m trajectories. At that point structure-factor parameters and image scale factors can be adjusted using intensity information rather than detector placement alone; quantitative relative-intensity claims require documented normalization and direct arrays.

For PbI_2, the ordered workflow is not discarded. It provides the geometry, projection, and mosaic baseline needed before diffuse scattering can be interpreted. The additional stacking-disorder stage asks whether a physically defined layer-correlation model accounts for the remaining diffuse intensity along fixed-Q_R cylinders. This keeps the PbI_2 result tied to the same detector-space framework while making clear which extra parameters belong specifically to stacking faults.

Implementation details, including lookup tables, Monte Carlo sampling, sub-pixel remapping, parameter-correlation plots, and the propagated uncertainty from 2θ–ϕ and Q transformations, should be documented in the supporting information. The main-text role of the workflow is to make the logic of the refinement reproducible: determine instrumental geometry first, determine sample orientation second, refine mosaic/profile effects third, and only then interpret ordered or diffuse intensity differences as structural information.

5 Results: diffuse scattering and stacking disorder in PbI_2

Figure 8 shows a CVD-grown PbI_2 film and a crop spanning the $m = 0$ and $m = 1$ Bragg rods. The $m = 1$ rod contains measurable intensity between the integer Bragg maxima, whereas the $m = 0$ rod is dominated by sharp peaks and finite-thickness oscillations. This rod-dependent redistribution of intensity is the experimental motivation for the stacking-disorder model. In a two-dimensional powder, random in-plane rotation maps each in-plane reflection family onto a reciprocal-space cylinder at fixed Q_R. Perfect stacking concentrates intensity at discrete Q_z values on a cylinder; loss of stacking coherence redistributes part of that intensity along the same cylinder.

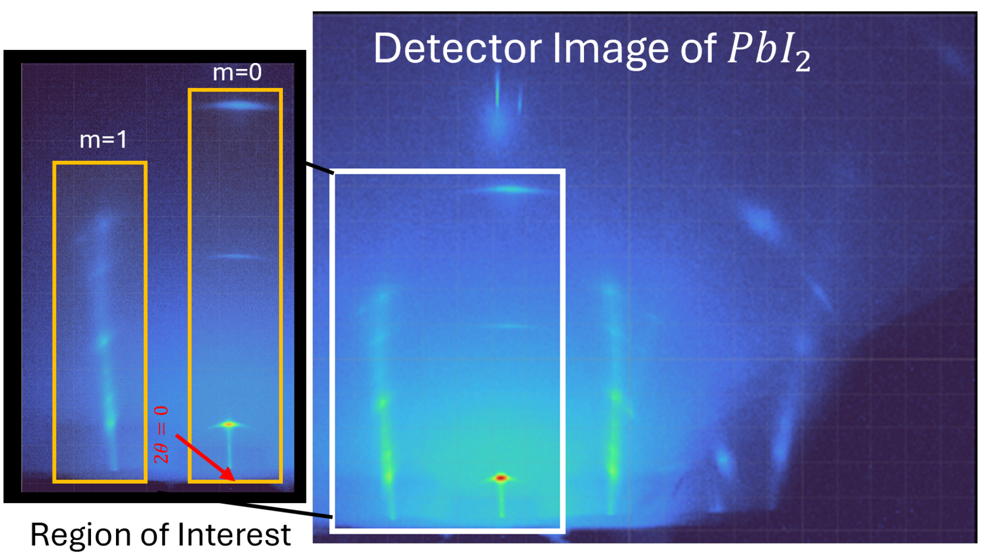


Figure 8: Measured PbI_2 diffraction image (right) and an enlarged region of interest (left). The direct-beam position defines the vertical $m = 0$ trajectory. Diffuse intensity is visible between the ordered maxima on the neighboring $m = 1$ rod, while the $m = 0$ profile remains insensitive to the lateral registry shifts that generate stacking contrast on other rods.

5.1 Direction-resolved transition-matrix model

The structural unit is one I–Pb–I trilayer. Its state is written $\alpha = (r, \sigma)$, where $r \in \{0, 1, 2\}$ denotes one of three basal registries and $\sigma \in \{+, -\}$ denotes the two trilayer orientations. The six states are therefore

$$\mathcal{S} = [0F_+, 1F_+, 2F_+, 0F_-, 1F_-, 2F_-] \,. \tag{19}$$

For a fixed m rod, a forward registry step $\mathbf{s} = (1/3, 2/3)$ contributes

$$\omega_{hk} = \exp[2\pi i(hs_x + ks_y)] = \exp\left[\frac{2\pi i}{3}(h + 2k)\right], \tag{20}$$

while the two local orientations carry form factors $F_+(h, k, L)$ and $F_-(h, k, L)$.

Adjacent trilayers obey the first-order interface law

$$\boldsymbol{\theta} = (a, b_+, b_-, d_+, d_-), \qquad a + b_+ + b_- + d_+ + d_- = 1. \tag{21}$$

Here a denotes no registry or orientation change. b_+ and b_- are forward and backward registry shifts without an orientation change. d_+ and d_- are the two handed shift-plus-orientation-change events. A change of orientation without a registry shift is excluded. The selected deterministic limits are $a = 1$ for 2H, $d_+ = 1$ for 4H, and $b_+ = 1$ for 6H. The opposite registry directions generate their symmetry-related handed counterparts. Figure 9 renders these cycles as projected I–Pb–I stacks. The sequences are read from bottom to top: $0F_+ \to 0F_+$ for 2H, $0F_+ \to 1F_- \to 0F_+$ for 4H, and $0F_+ \to 1F_+ \to 2F_+ \to 0F_+$ for 6H. These parent structures and PbI_2 polytypism are established experimentally.[32, 33]

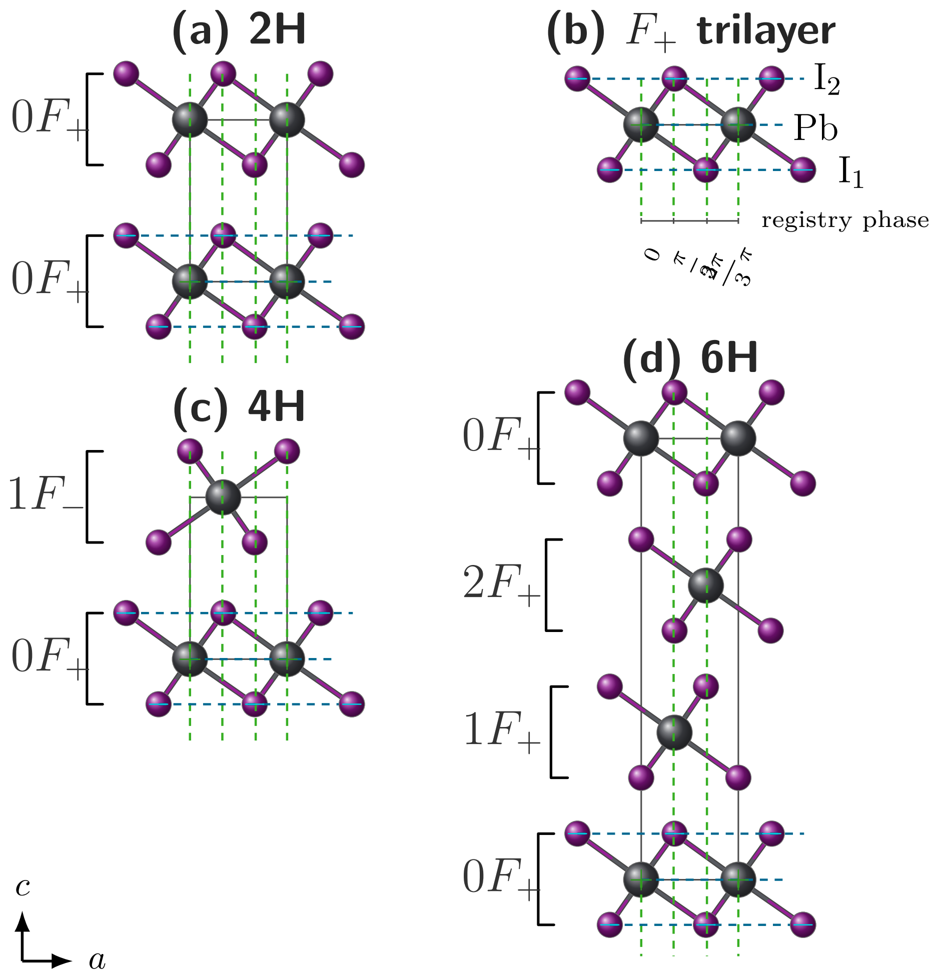


Figure 9: Projected I–Pb–I trilayers for the deterministic PbI_2 parent cycles used in the transition model. (a) 2H repeats $0F_+$ under $a = 1$. (b) An isolated F_+ trilayer shows the Pb and I atomic planes used to define the local orientation. (c) 4H alternates $0F_+$ and $1F_-$ under $d_+ = 1$. (d) 6H advances $0F_+ \to 1F_+ \to 2F_+ \to 0F_+$ under $b_+ = 1$. The c axis points upward, so each stack is read from bottom to top. In a state label rF_σ, $r \in \{0, 1, 2\}$ is the basal registry and $\sigma \in \{+, -\}$ is the trilayer orientation. Violet and gray spheres denote I and Pb, green vertical guides mark lateral registry, blue horizontal guides mark atomic planes, and brackets group each I–Pb–I trilayer. Signs are shown on the reference F_+ trilayer.

The general layer-correlation treatment follows the established transition-matrix theory of one-dimensionally disordered crystals, whereas the six states and allowed PbI_2 interface events above are the material-specific construction used here.[16, 34–36] Because the registry labels enter a given rod only through 1, ω_{hk}, and ω_{hk}^{-1}, the full six-state matrix reduces exactly to the following phase-weighted matrix in the orientation basis:

$$M(\omega) = \begin{pmatrix} a + b_+\omega + b_-\omega^{-1} & d_+\omega + d_-\omega^{-1} \\ d_+\omega^{-1} + d_-\omega & a + b_+\omega + b_-\omega^{-1} \end{pmatrix}. \tag{22}$$

The probability matrix $P = M(1)$ propagates the orientation populations, while $M(\omega)^\Delta$ sums the probabilities and registry phases of all paths connecting two trilayers separated by Δ interfaces. The complete construction and its reduction are given in Supplementary Note S3..

For each parent-rich population $j \in \{2H, 4H, 6H\}$, the disorder parameter has a direct probabilistic meaning:

$$p_j(\text{parent event}) = 1 - \epsilon_j, \qquad p_j(\text{each of the four alternative events}) = \frac{\epsilon_j}{4}. \tag{23}$$

For 2H, 4H, and 6H the parent event is a, d_+, and b_+, respectively. Written in the order (a, b_+, b_-, d_+, d_-), the five probabilities in Eq. (23) form $\boldsymbol{\theta}_j$ and are inserted directly into $M_j(\omega)$ and $P_j = M_j(1)$. Thus ϵ_j changes the matrix powers that determine how much phase coherence remains between increasingly distant trilayers.

For one definite sequence $s_n = (r_n, \sigma_n)$, the layer amplitude is $g_{s_n} = \omega^{r_n} F_{\sigma_n}$ and the coherent stack amplitude is $A_N(L) = \sum_{n=0}^{N-1} \xi(L)^n g_{s_n}$, where $\xi(L) = \exp(iq_z d)$ is the vertical phase between neighboring trilayer centers. Expanding $\langle |A_N|^2 \rangle$ leaves two familiar contributions: the intensity scattered by each trilayer individually and the interference between distinct trilayer pairs.

For population j, define the orientation population of trilayer n as

$$\boldsymbol{\pi}_{j,n} = \left(p^+_{j,n}, p^-_{j,n}\right) = \boldsymbol{\pi}_{j,0} P_j^n, \qquad p^+_{j,n} + p^-_{j,n} = 1. \tag{24}$$

Here j identifies the 2H-, 4H-, or 6H-rich population and n identifies a trilayer within the finite stack. The quantities $p^+_{j,n}$ and $p^-_{j,n}$ are ordinary probabilities that trilayer n has orientation F_+ or F_-, respectively; they are neither scattering amplitudes nor the sample weights w_j used below.

The disorder-averaged interference amplitude for trilayers n and $n + \Delta$ is

$$K^{(j)}_{n,\Delta} = \begin{pmatrix} p^+_{j,n} & p^-_{j,n} \end{pmatrix} \begin{pmatrix} F_+^* & 0 \\ 0 & F_-^* \end{pmatrix} M_j(\omega)^\Delta \begin{pmatrix} F_+ \\ F_- \end{pmatrix}. \tag{25}$$

The finite-stack intensity is then written as "self scattering plus pair interference":

$$I_j(L) = I_{j,\text{self}}(L) + I_{j,\text{pair}}(L), \tag{26a}$$

$$I_{j,\text{self}}(L) = \frac{1}{N} \sum_{n=0}^{N-1} \left[p^+_{j,n} |F_+|^2 + p^-_{j,n} |F_-|^2 \right], \tag{26b}$$

$$I_{j,\text{pair}}(L) = \frac{2}{N} \operatorname{Re} \sum_{\Delta=1}^{N-1} \xi(L)^\Delta \sum_{n=0}^{N-1-\Delta} K^{(j)}_{n,\Delta}. \tag{26c}$$

The self term averages the isolated-trilayer intensity over the two possible orientations. The pair term groups all trilayer pairs by their separation Δ: there are $N - \Delta$ pairs at that separation, and each carries the same vertical phase $\xi(L)^\Delta$. In Eq. (25), the expression can be read from right to left. The final column supplies the scattering amplitude of the later trilayer, $M_j(\omega)^{\Delta}$ sums all allowed stacking paths and registry phases between the two layers, the diagonal matrix supplies the conjugated amplitude of the earlier trilayer, and $\boldsymbol{\pi}_{j,n}$ averages over whether that earlier trilayer is in the F_+ or F_- orientation. The factor $2\,\mathrm{Re}$ in Eq. (26c) includes the reverse-ordered conjugate pair without writing it separately. The complete derivation is given in Supplementary Note S3.

The 2H-, 4H-, and 6H-rich domains are taken to be mutually incoherent, so the three calculated intensities (not their amplitudes) are combined as

$$I_{\mathrm{calc}}(L) = S\left\{[w_{2H}I_{2H}(L) + w_{4H}I_{4H}(L) + w_{6H}I_{6H}(L)] \otimes R(L)\right\}, \qquad w_{2H} + w_{4H} + w_{6H} = 1. \tag{27}$$

Each $I_j(L)$ is first evaluated from Eq. (26) using its own ϵ_j. The weights then specify the fraction of the total calculated intensity assigned to each parent-rich population, the shared resolution function $R(L)$ broadens the combined profile, and S converts the normalized calculation to the measured intensity scale. Accordingly, ϵ_j is an effective per-interface disorder probability under the stated equal-alternative template, whereas w_j is an effective population weight.

For $h = k = 0$, $\omega_{hk} = 1$ and $F_+ = F_-$. Registry shifts and orientation changes then leave the scattering amplitude unchanged, so stacking faults are contrast-matched rather than absent. The oscillations on the $m = 0$ rod are therefore finite-thickness fringes. The same criterion applies to any other rod for which both the registry phase and the orientation form factors are indistinguishable.

5.2 Refinement

The disorder stage is applied only after the detector geometry, sample alignment, mosaic distribution, lattice constants, ordered PbI_2 structure factor, and profile resolution have been fixed by the upstream refinement. Measured and calculated detector images are projected through the same constant-Q_R rod masks and reported versus Q_z or the corresponding stacking coordinate L. No independent peak areas are introduced before the stacking model is evaluated, because doing so would absorb the diffuse between-peak intensity that the transition law is intended to explain.

The core disorder parameters are

$$\boldsymbol{\Theta}_{\mathrm{dis}} = (\epsilon_{2H}, \epsilon_{4H}, \epsilon_{6H}, w_{2H}, w_{4H}, w_{6H}, N), \qquad w_{2H} + w_{4H} + w_{6H} = 1, \tag{28}$$

so the three weights contain only two independent degrees of freedom. A stable refinement can first impose a shared disorder magnitude $\epsilon_{2H} = \epsilon_{4H} = \epsilon_{6H}$ and release the three values only when the retained fault-sensitive rods support phase-dependent broadening. The selected-profile residual is

$$r_j = \frac{I_j^{\mathrm{meas}} - I_j^{\mathrm{model}}(\boldsymbol{\Theta}_{\mathrm{dis}})}{\sigma_j}, \tag{29}$$

with any allowed global scale or low-order background terms declared separately. The fit is accepted only when it improves the selected diffuse profiles without degrading the ordered detector-space agreement established in the preceding stages.

Disorder Series Fitting

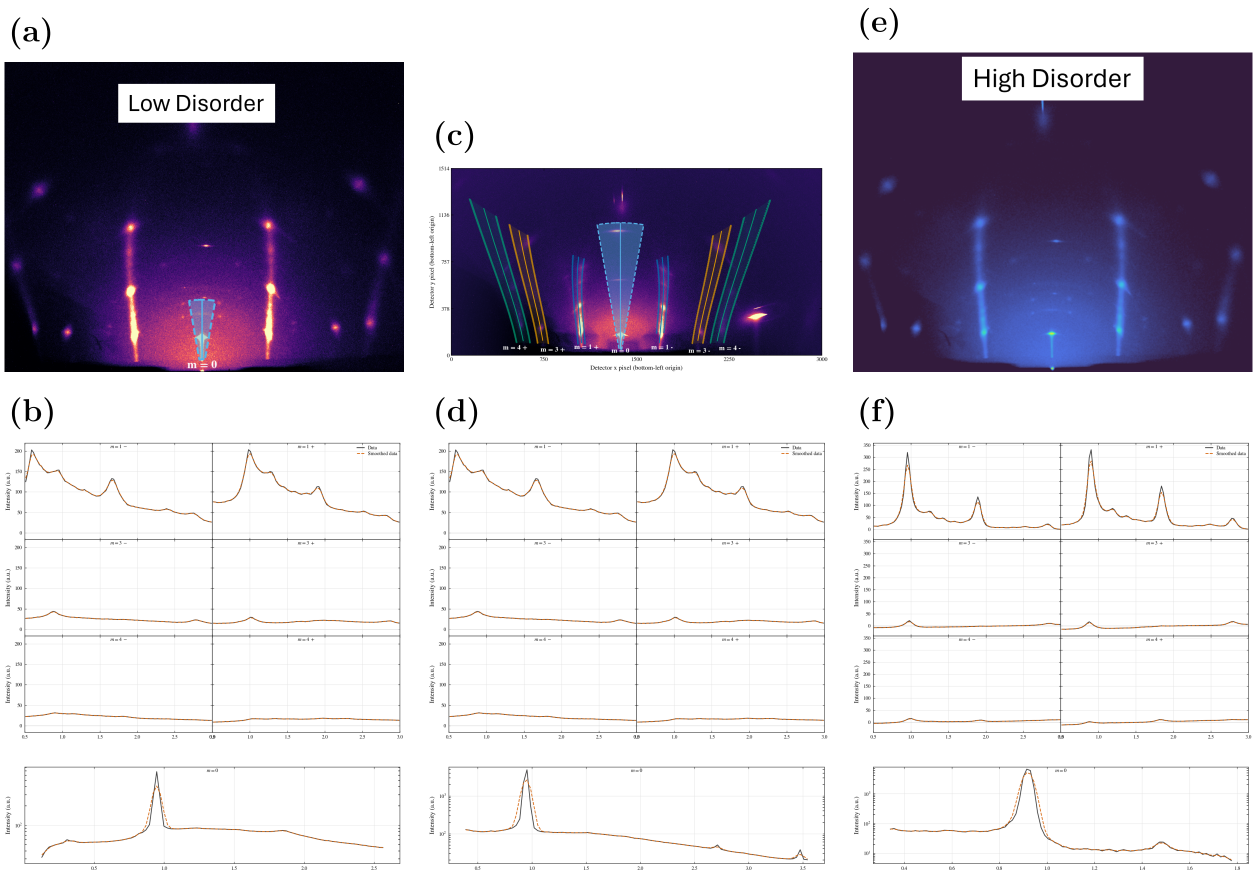


Figure 10: PbI_2 diffuse-rod comparison at $\theta_i = 4°$ for samples spanning increasing stacking disorder. The top row shows measured detector images and the constant-Q_R extraction regions. Solid curves mark the projected rod centerlines, shaded bands mark the integrated detector pixels, and dashed outlines mark the integration boundaries. The bottom row compares profiles obtained by applying the same projection to data and calculation; black solid curves are measured and orange dashed curves are calculated. The series tests whether one detector-space geometry and ordered baseline, augmented by the transition-matrix stacking model, reproduce the progressive transfer of intensity from ordered maxima into the diffuse rod signal.

Figure 10 shows the visible progression from sharp ordered maxima to stronger between-peak intensity. The absence of comparable diffuse contrast on $m = 0$ does not imply that those scattering volumes contain no faults; it follows from the phase-insensitive condition described above. Likewise, changes in the visibility of higher order $m = 0$ peaks are assigned to the growth-dependent mosaic and resolution state rather than to lateral-registry disorder.

The ordered baseline and the transition-model parameterization are summarized in Table 2. Numerical values from the superseded scalar two-population fit are not carried into the new model because they do not map uniquely onto three population-specific ϵ_j values and three normalized weights.

Table 2: PbI_2 ordered-baseline parameters and stacking-disorder parameterization for the selected-rod comparison. Numerical transition-matrix parameters must be reported from a refit using Eqs. (23)–(27); values from the superseded scalar model are not reused.

Parameter class	Fitted value or model quantity	Literature comparison	Role in the selected-rod model
Lattice and coordinates	$a = 4.5583$ Å, $c = 6.985$ Å; Pb at $(0, 0, 0)$; I1 at $(1/3, 2/3, 0.2677)$ and I2 at $(2/3, 1/3, 0.7323)$.	Lattice constants remain within the 2H single-crystal structural spread, and the iodine coordinate follows the corrected $z \simeq 0.268$ value.[32, 33]	Fixes the ordered Bragg positions and the $\lvert F_{hk}(L)\rvert^2$ envelope before stacking disorder is introduced.
Occupancies	A weakly floating test returned $o_{\mathrm{Pb}} = 0.999$, $o_{\mathrm{I1}} = 1.001$, and $o_{\mathrm{I2}} = 1.000$; the ordered baseline was then evaluated with full occupancies.	The clean ordered model uses full occupancies; the density-deficient occupancy reported for a Palosz-type sample is sample-specific and was not used as the default model.[33, 37]	Keeps the PbI_2 average structure stoichiometric while the diffuse rod intensity is assigned to stacking correlations.
Directional ADPs	$U_R/U_z = 0.0105/0.0185$ for Pb and $0.016/0.024$ Å^2 for I1 and I2.	Site-resolved Pb and I ADPs are not widely or consistently reported for direct transfer; the qualitative inequality $U_z(\mathrm{Pb}) > U_R(\mathrm{Pb})$ is retained.[32, 38]	Moderates relative ordered intensities and is interpreted as a sample-specific displacement refinement rather than a universal PbI_2 displacement constant.
Transition-matrix stacking disorder	ϵ_{2H}, ϵ_{4H}, ϵ_{6H}; normalized weights w_{2H}, w_{4H}, w_{6H}; and finite stack length N. Final numerical values are pending the transition-matrix refit.	These are effective selected-rod descriptors under the stated parent rules and equal-alternative fault templates, not universal PbI_2 constants.	Controls the redistribution of the ordered rod envelope into broadened maxima, finite-thickness fringes, and diffuse between-peak intensity.

The physical interpretation is therefore deliberately conditional on the model. Each ϵ_j measures the probability of departing from the selected parent event within population j, the w_j describe the incoherent population mixture, and N controls the finite-stack interference. Numerical values should be inserted only after the experimental profiles have been refit with this parameterization.

6 Discussion and conclusion

The ordered Bi_2Se_3 and Bi_2Te_3 test cases compare measured and calculated detector features after geometry, finite beam phase space, sample alignment, mosaicity, resolution, and ordered structure-factor weighting are applied. Measured and calculated images are projected along the same fixed-m trajectories. Within the fitted data set, the selected profiles agree in their principal peak positions and apparent line shapes, including features that would be difficult to interpret from a one-dimensional reduction alone.

Keeping the comparison in detector space is essential because the constraining intensity is distributed among off-specular arcs, cap-like $m = 0$ features, weak off-condition specular peaks, and bright low-L specular-region intensity that a conventional one-dimensional reduction would obscure. Applying the same projection to data and calculation makes the comparison direct, but it does not remove correlations among mosaicity, bandwidth, divergence, alignment, background, and optical response. The off-condition $m = 0$ peaks motivate a weak broad mosaic component; the present in-sample overlays do not establish that this component is uniquely Lorentzian or necessary relative to a re-optimized Gaussian-only model.

This distinction defines the relationship to detector-space indexing software. Programs such as indexGIXS calculate and overlay allowed reflection positions, including refraction-corrected spot locations, and are therefore the appropriate prior art for the geometric indexing stage.[29] The contribution here is the subsequent detector-space intensity/profile calculation: the same geometry is used to compare measured and calculated feature locations and profile shapes after beam phase space, mosaicity, bandwidth, resolution, optical attenuation, structure-factor weighting, and disorder terms are applied.

The PbI_2 section outlines how the same framework can be extended from ordered Bragg line-shape comparisons to diffuse scattering from stacking disorder. In that extension, the detector geometry, beam phase space, mosaicity, and projection procedure remain the same, but the reciprocal-space intensity is generated from a stacking-fault correlation model rather than from a perfectly periodic ordered stack. This separation is important: diffuse intensity should be interpreted as stacking information only after geometry and mosaicity have been constrained by the ordered-film workflow.

The practical refinement workflow is therefore part of the scientific argument. Instrumental parameters are determined first, sample alignment second, mosaic and profile parameters third, and ordered or diffuse intensity parameters last. This ordering reduces parameter compensation and makes the final data/model comparison interpretable. The manuscript follows the same logic: show the observation, explain the physical inference and model in reciprocal-space terms, and then place measured and calculated quantities on matched axes.

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